# Sudan rainfall extremes

Observed thresholds and ensemble forecast signals

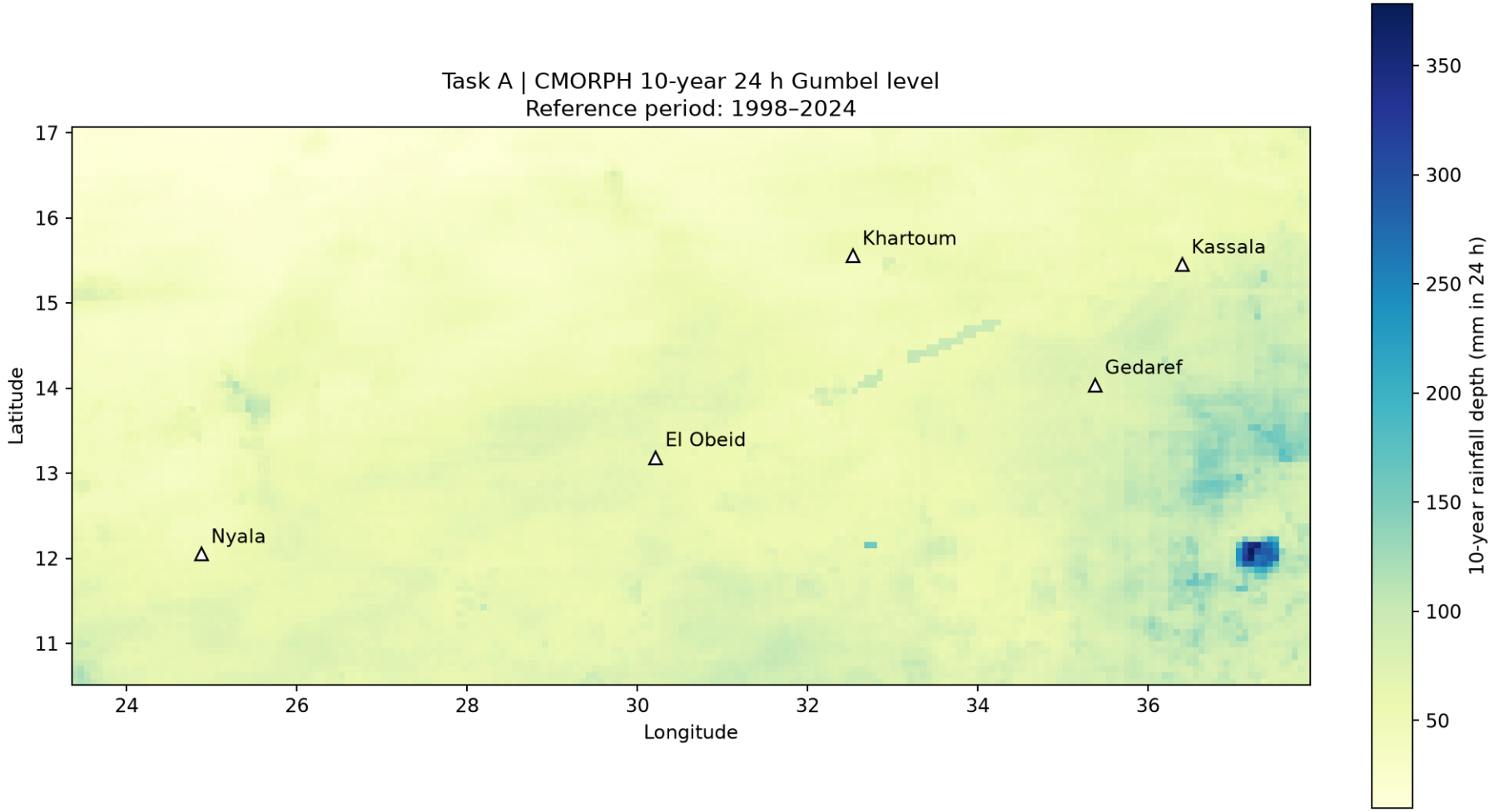
Five towns from grid.json

Tasks A–D

CMORPH: 1998–2024
ECMWF ensemble: June–August 2025

# Five towns and the observed rainfall reference

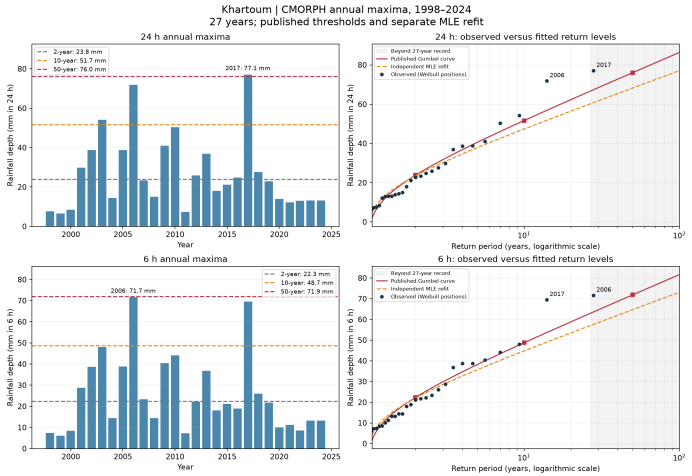
Task A regional 10-year level, in mm accumulated over 24 h

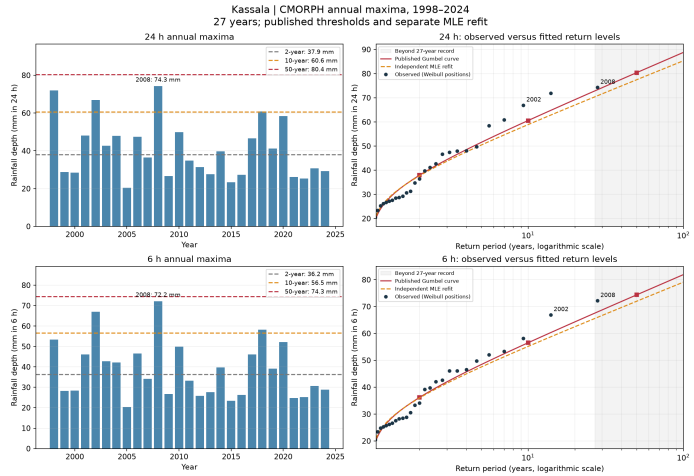


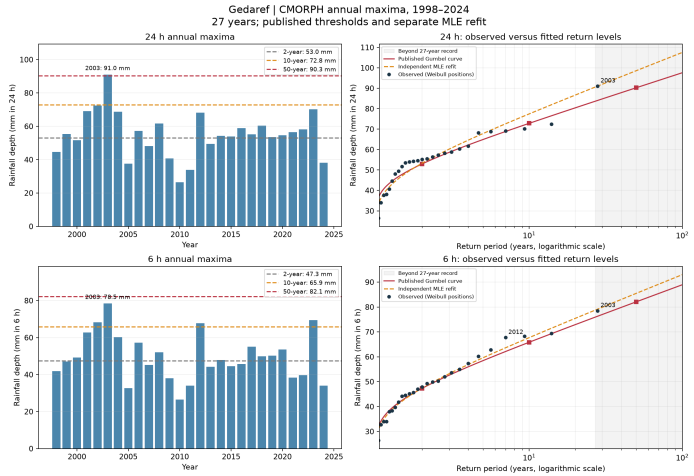

CMORPH annual maxima: 1998–2024, 27 years. Five town coordinates from grid.json.

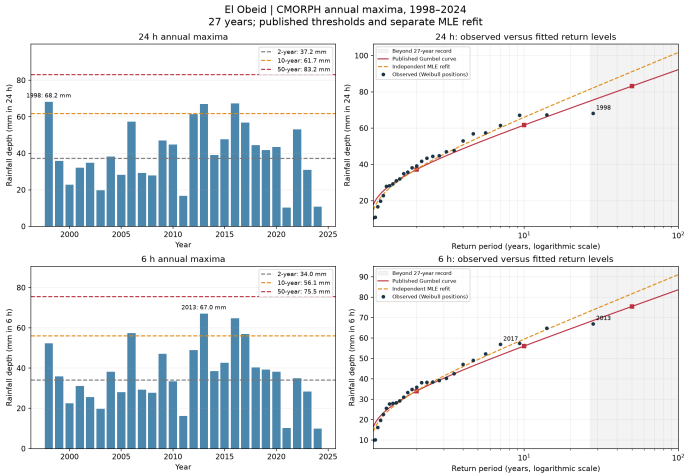
# A: observed extremes at every town

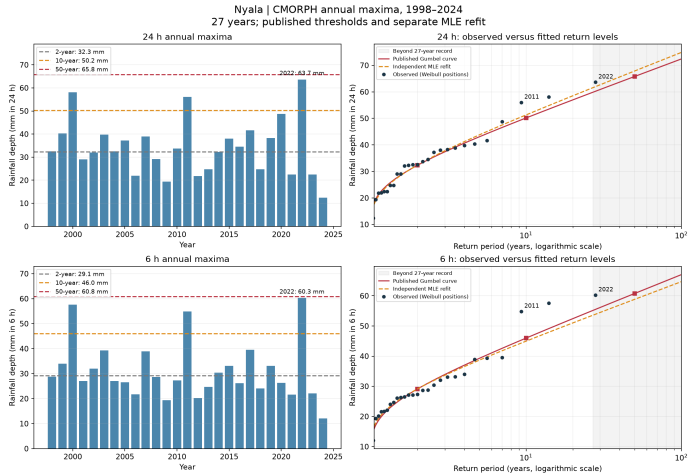
## Annual maxima and fitted return levels for 24 h and 6 h




The plots distinguish published Gumbel levels from an independent MLE refit. Large departures are descriptive evidence.

# A: local return levels and largest-year sensitivity

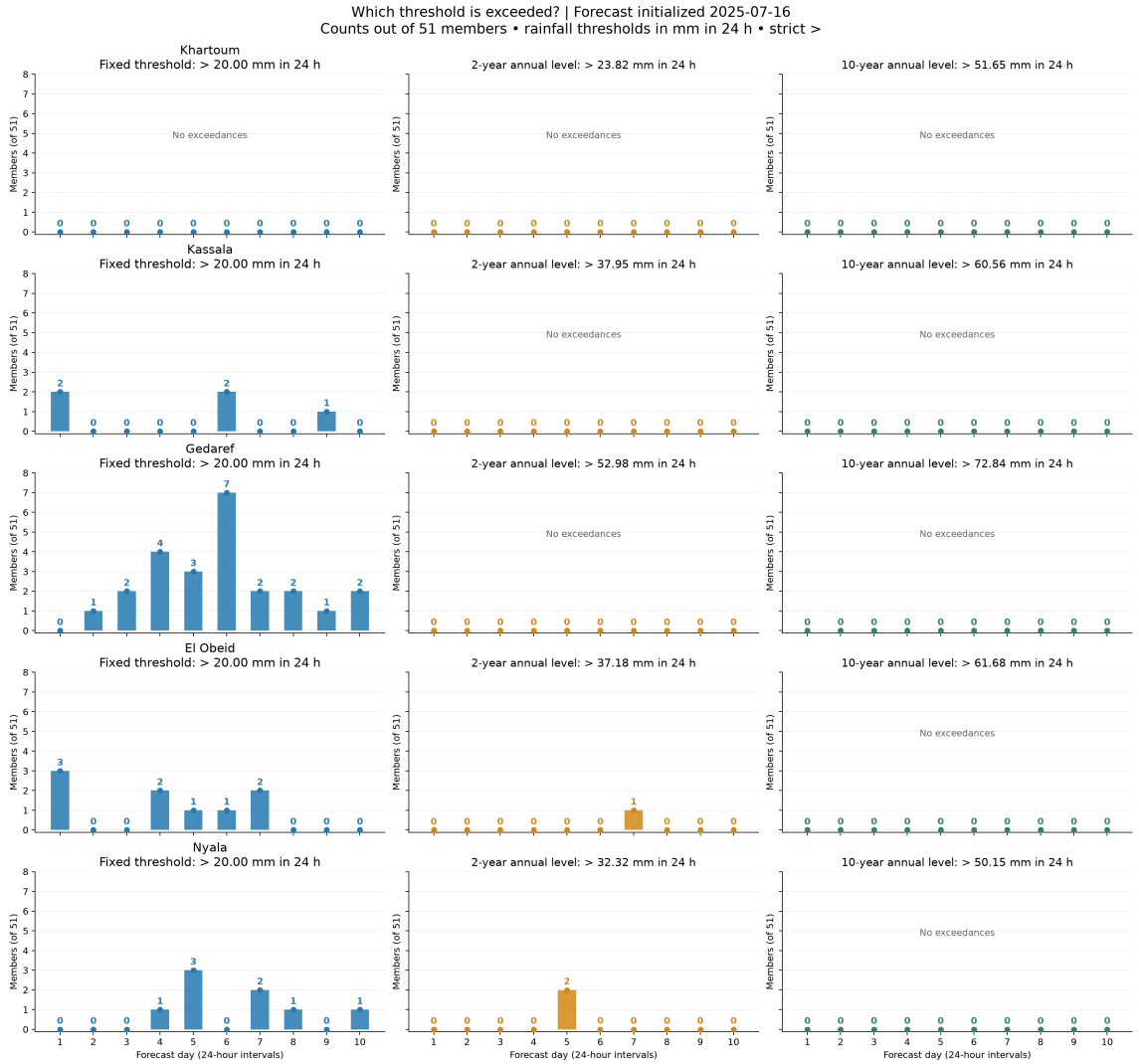
Published 24 h rainfall depths in mm

| Town | 2-year | 10-year | 50-year |
|---|---|---|---|
| Khartoum | 23.82 | 51.65 | 76.05 |
| Kassala | 37.95 | 60.56 | 80.39 |
| Gedaref | 52.98 | 72.84 | 90.25 |
| El Obeid | 37.18 | 61.68 | 83.16 |
| Nyala | 32.32 | 50.15 | 65.79 |

Removing Khartoum’s 2017 maximum lowers its MLE 50-year level from 68.23 to 62.05 mm (−9.05%). Both sensitivity fits use MLE. Fifty-year levels extrapolate beyond 27 observed years.

# B: three thresholds across five towns

16 July 2025 start, ten daily intervals, 51 members



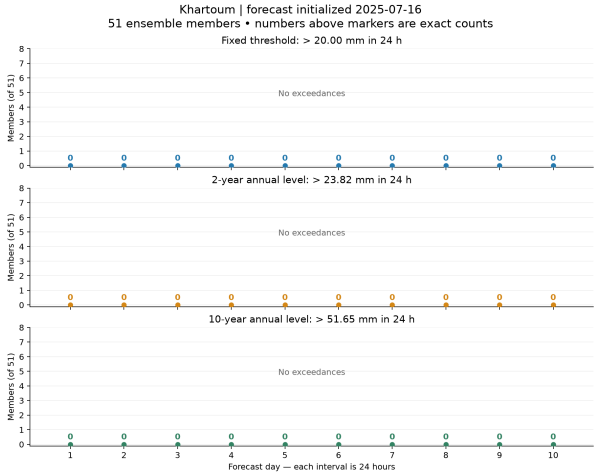

Fixed threshold: 20 mm in 24 h

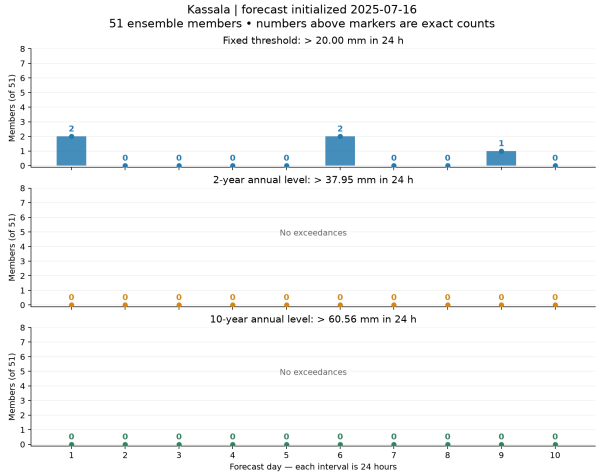
Local 2-year and 10-year levels come from Task A.

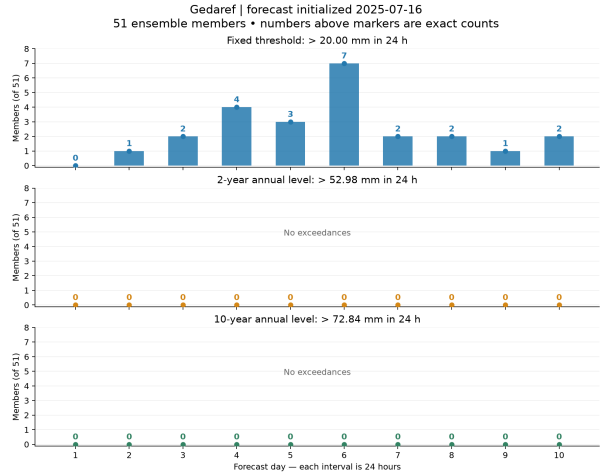
El Obeid and Nyala alone have 2-year exceedances. No town exceeds a 10-year level.

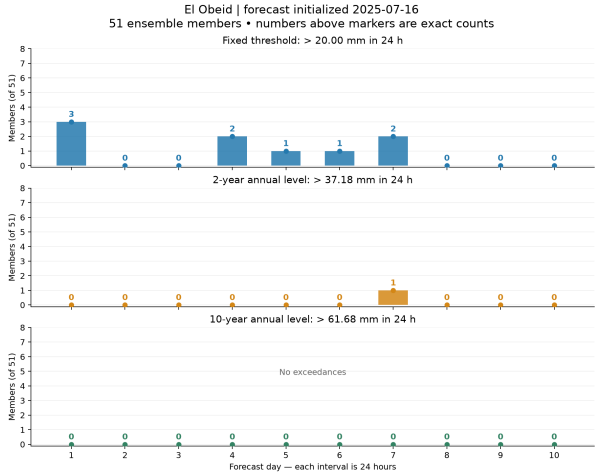
# B: daily member counts at each town

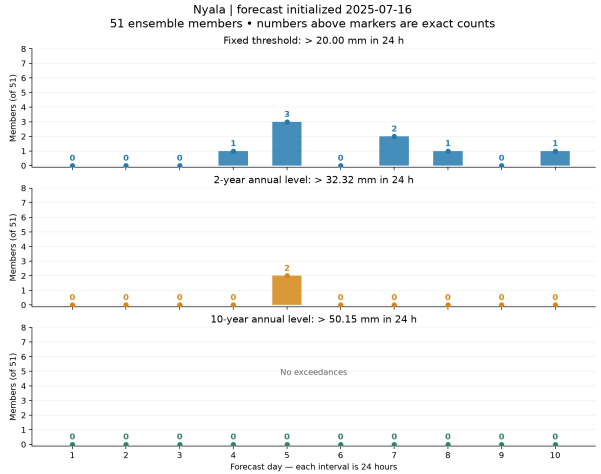
## Each figure separates the fixed 20 mm, 2-year and 10-year thresholds




Daily counts are out of 51 members. Rainfall thresholds are mm in 24 h. Exact zero labels distinguish no exceedance.

# B: threshold sensitivity and xclim counts

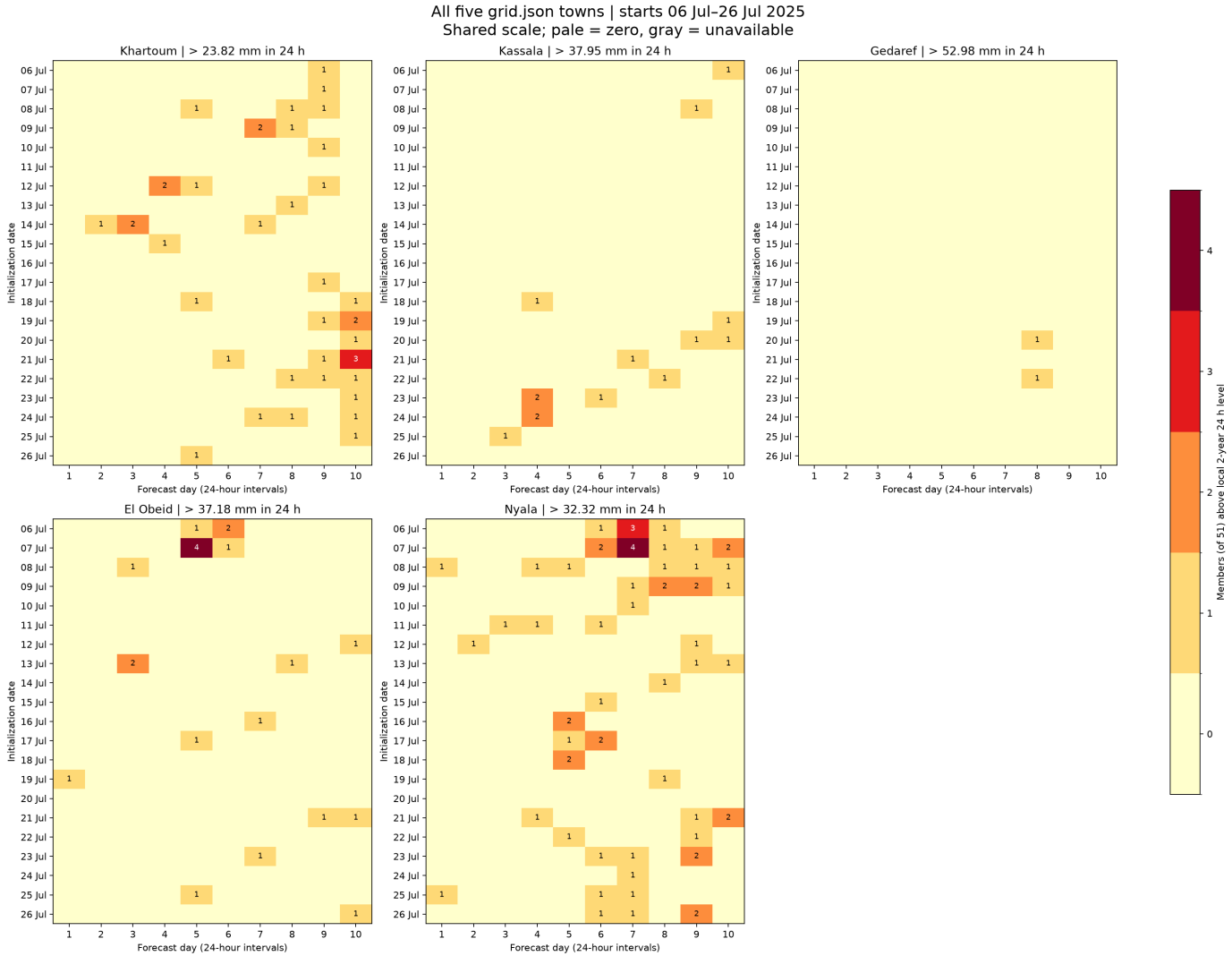
First three columns count member-days out of 510 per town. Last column: distinct members at ≥20 mm.

| Town | 20 mm | 2-year | 10-year | ≥1 heavy day |
|---|---|---|---|---|
| Khartoum | 0 | 0 | 0 | 0/51 |
| Kassala | 5 | 0 | 0 | 5/51 |
| Gedaref | 24 | 0 | 0 | 22/51 |
| El Obeid | 9 | 1 | 0 | 7/51 |
| Nyala | 8 | 2 | 0 | 7/51 |

xclim uses ≥, plots use >. A 2-year annual level implies 50% annual exceedance. The illustrative 0.190% daily probability assumes independent, identical days.

# C: forecast signals across 21 starts

## 6–26 July 2025, ten daily leads at every town



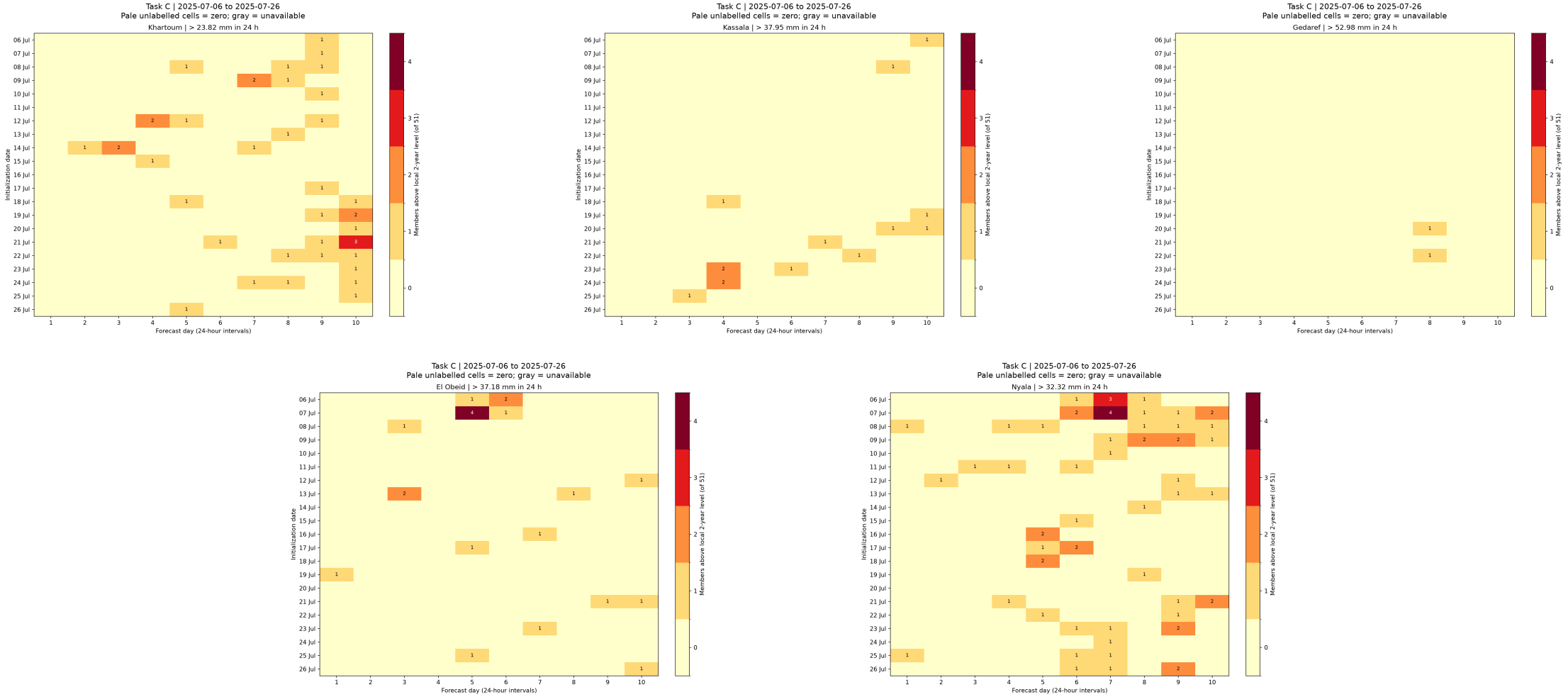

Signal: at least one member above the local 2-year 24 h level.

Nyala: signals in 20/21 starts
Gedaref: signals in 2/21 starts

A one-member signal is only 1.96% of the ensemble.

# C: complete daily signal histories

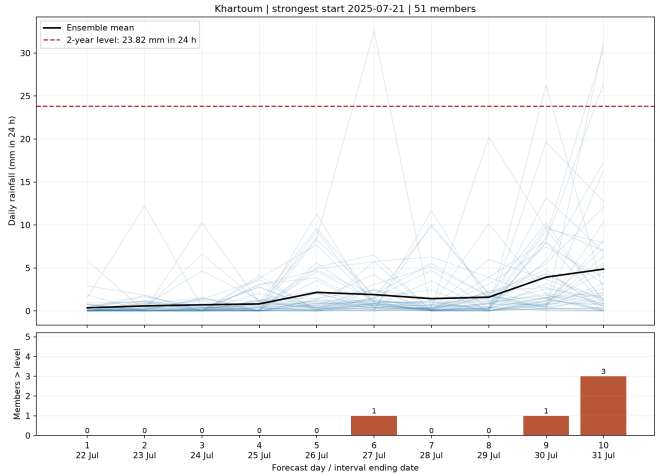
## Rows are initialization dates, columns are forecast days

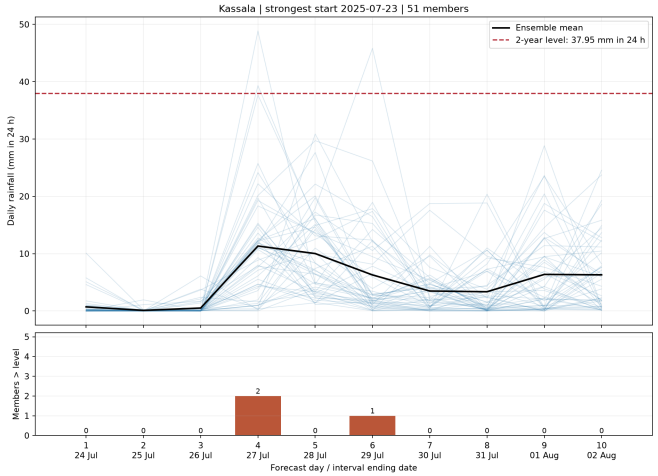


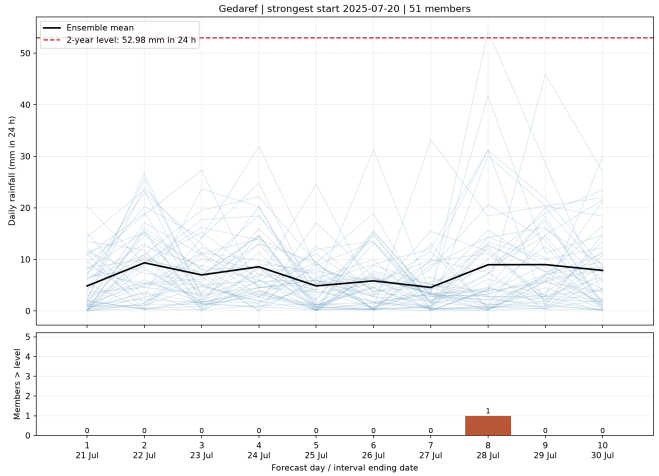

Shared member-count scale across towns. Local rainfall thresholds differ. All 1,050 town/start/day cells are available.

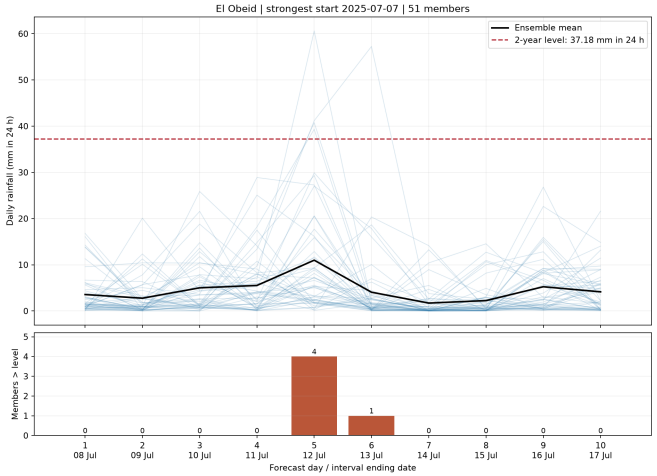
# C: the strongest start at each town

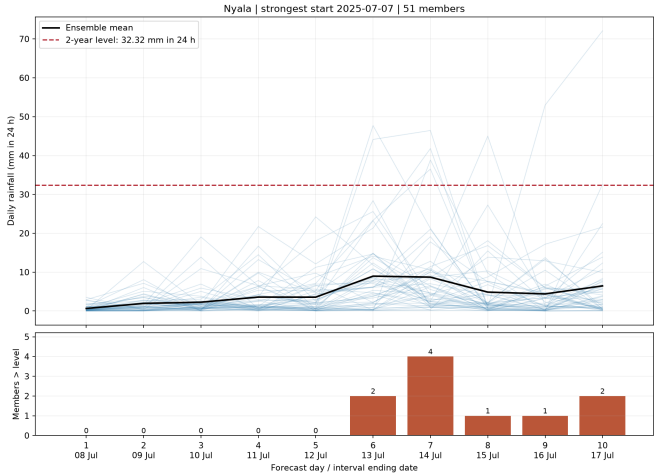
## Daily rainfall plumes, ensemble mean, local threshold and member counts




Strongest start maximizes peak count, then summed member-day counts, then earliest date. Plume rainfall units: mm in 24 h.

# C: Nyala has the most persistent weak signal

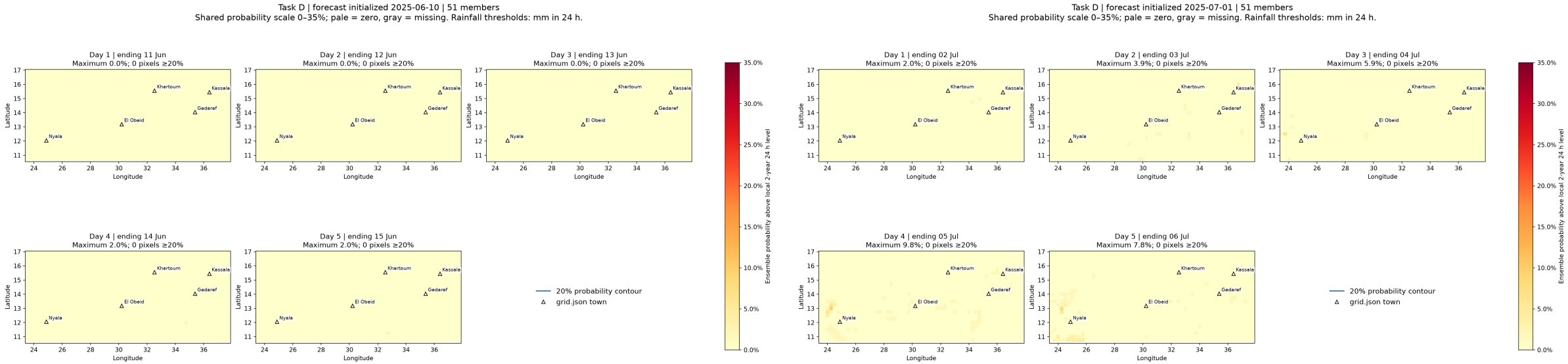
Streaks require consecutive starts signaling the same target interval

| Town | Peak | Lead days | Signal streak | Streak target |
|---|---|---|---|---|
| Khartoum | 3/51 | 10 | 3 starts | 2025-07-16 |
| Kassala | 2/51 | 10 | 2 starts | 2025-07-28 |
| Gedaref | 1/51 | 8 | 1 start | 2025-07-28 |
| El Obeid | 4/51 | 10 | 2 starts | 2025-07-12 |
| Nyala | 4/51 | 10 | 5 starts | 2025-07-17 |

Nyala’s starts on 7–11 July target the interval ending 17 July. Four towns tie at the ten-day horizon, targeting different dates. No town reaches five members above the threshold.

# D: probability maps for the early seasonal samples

## Local 2-year 24 h thresholds and all five grid.json towns



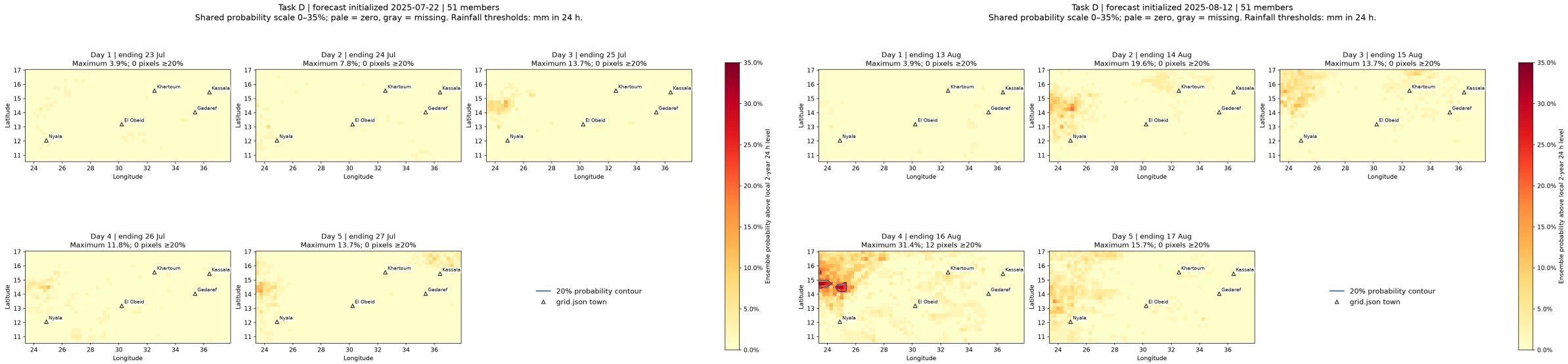




June 10 peak: 1.96%. July 1 peak: 9.80%. No sampled pixel reaches 20% for these two starts.

# D: the August sample contains the regional hotspot

## Probability decision: ≥20%, requiring at least 11 of 51 members



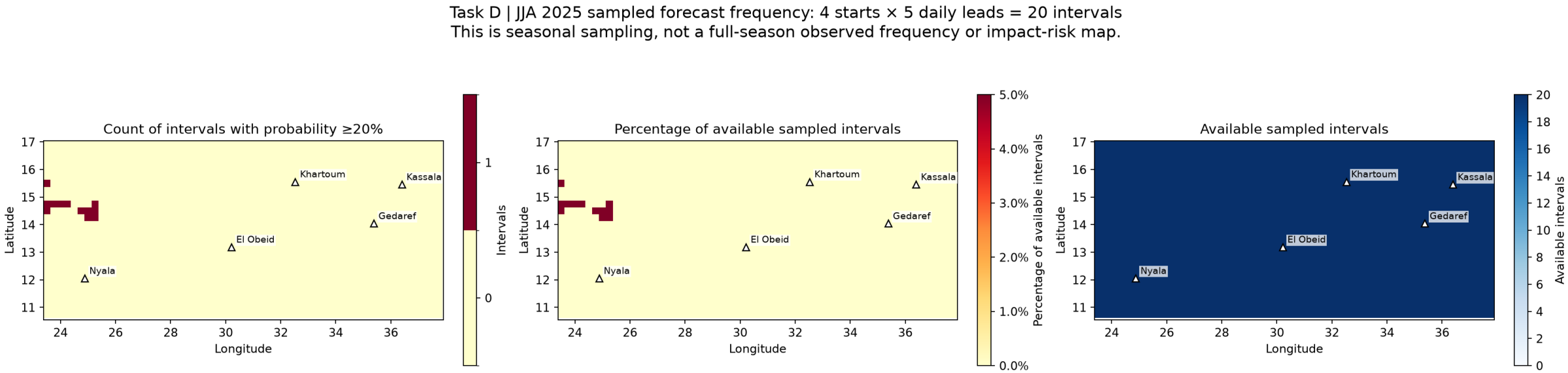




August 12, day 4: 12 pixels reach 20%, with a peak of 16/51 = 31.37%. The target interval ends 16 August.

# D: sampled frequency and town results

## Four starts × five daily leads = 20 sampled forecast intervals




Twelve regional pixels reach 20% once each. Maximum frequency: 1/20 = 5%. All five town pixels remain below 20%.

# Conclusions and the cost of map reads

Pan chunks contain a broad map, all 51 members, and one start/lead

| Start | Cloud read | Chunks |
|---|---|---|
| 2025-06-10 | 37.43 s | 6 |
| 2025-07-01 | 19.49 s | 6 |
| 2025-07-22 | 21.14 s | 6 |
| 2025-08-12 | 34.88 s | 6 |

Six cumulative endpoints give five daily maps. Threshold read: 10.84 s, one chunk. Local thresholds shape weak signals. Observed 2025 rainfall and impacts are needed to assess warning skill.